Summary in Graph

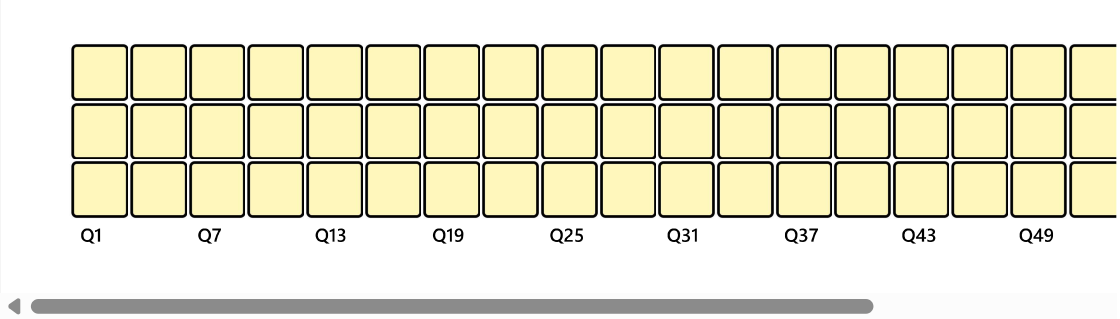
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 9)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (12) out of 119 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

The number $25^{64} \cdot 64^{25}$ is the square of a positive integer N. In decimal representation, the sum of the digits of N is

A. 7
B. 14
C. 21

D. 28

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 03sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Francesca uses 100 grams of lemon juice, 100 grams of sugar, and 400 grams of water to make lemonade. There are 25 calories in 100 grams of lemon juice and 386 calories in 100 grams of sugar. Water contains no calories. How many calories are in 200 grams of her lemonade?

A. 129
B. 137
C. 174
D. 233

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

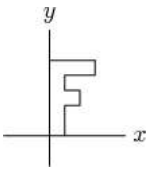
Two years ago Pete was three times as old as his cousin Claire. Two years before that, Pete was four times as old as Claire. In how many years will the ratio of their ages be $2:1$?

A. 2
B. 4
C. 5
D. 6

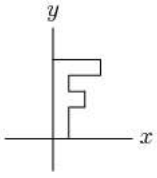
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

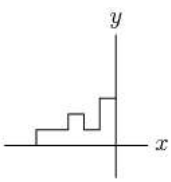
The letter F shown below is rotated 90° clockwise around the origin, then reflected in the y-axis, and then rotated a half turn around the origin. What is the final image?



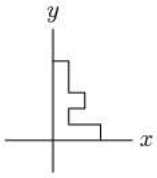
A.



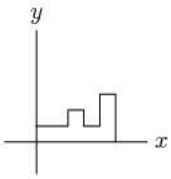
B.



C.



D.



Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

A. was wishing
B. is wishing
C. wished
D. had wished

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Johann has 64 fair coins. He flips all the coins. Any coin that lands on tails is tossed again. Coins that land on tails on the second toss are tossed a third time. What is the expected number of coins that are now heads?

A. 32
B. 40
C. 48
D. 56

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Three generations of the Wen family are going to the movies, two from each generation. The two members of the youngest generation receive a 50% discount as children. The two members of the oldest generation receive a 25% discount as senior citizens. The two members of the middle generation receive no discount. Grandfather Wen, whose senior ticket costs $\$6.00$, is paying for everyone. How many dollars must he pay?

A. 34
B. 36
C. 42
D. 46

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Bernardo and Silvia play the following game. An integer between 0 and 999 inclusive is selected and given to Bernardo. Whenever Bernardo receives a number, he doubles it and passes the result to Silvia. Whenever Silvia receives a number, she adds 50 to it and passes the result to Bernardo. The winner is the last person who produces a number less than 1000 . Let N be the smallest initial number that results in a win for Bernardo. What is the sum of the digits of N ?

A. 7
B. 8
C. 9
D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

For $x > 0$, for which range of values of α is the following inequality true?

$$x \log_e(x) \geq x - \alpha$$

A. $\alpha \geq 1/2$
B. $\alpha \geq 0$
C. $\alpha \leq 2$
D. $\alpha \geq 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Out of the following 4 sentences, select the most suitable sentence with respect to grammar and usage:

A. Since the report lacked needed information, it was of no use to them.
B. The report was useless to them because there were no needed information in it.
C. Since the report did not contain the needed information, it was not real useful to them.
D. Since the report lacked needed information, it would not had been useful to them.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

In class, we showed that the expected squared error can be decomposed into several important terms:

$$\mathbb{E}\left[\left(Y - f_{\hat{\theta}}(x)\right)^2\right] = \sigma^2 + \left(h(x) - \mathbb{E}\left[f_{\hat{\theta}}(x)\right]\right)^2 + \mathbb{E}\left[\left(\mathbb{E}\left[f_{\hat{\theta}}(x)\right] - f_{\hat{\theta}}(x)\right)^2\right].$$

For which of the following reasons are we taking an expectation? In other words, what are the sources of randomness that we are considering in the derivation of the bias-variance tradeoff?

A. We chose arbitrary features when doing feature engineering.
B. We drew random samples from some larger population when we built our training set.
C. There is some noise in the underlying process that generates our observations $\mathbf{Y}$ from our features.

D. Our x values could have had missing or erroneous data, e.g. participants misreading a question on a survey.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following apply to linear discriminant analysis?

A. You calculate the sample mean for each class
B. You calculate the sample covariance matrix using the mean of all the data points
C. It approximates the Bayes decision rule
D. The model produced by LDA is never the same as the model produced by QDA

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Given a dataset $\mathrm{D}=\{(x_i, y_i)\}, x_i \in R^k, y_i \in=\{-1,+1\}, 1 \leq i \leq N$.
Which of the following is the Hinge Loss of the example (x_i, y_i) ?

A. $\max\left(0,-y_i \mathbf{w}^T \mathbf{x}_i\right)$
B. $\min\left(0,1-y_i \mathbf{w}^T \mathbf{x}_i\right)$
C. $\max\left(0,1-y_i \mathbf{w}^T \mathbf{x}_i\right)$
D. $\max\left(0, y_i \mathbf{w}^T \mathbf{x}_i\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Assume you have a set of independent observations from a normally distributed population: $2,5,7,4$. If the mean (μ) of the population is known to be 4 , calculate the Maximum Likelihood Estimation (MLE) of the variance $\left(\sigma^2\right)$.

A. $\sigma^2=3.0$
B. $\sigma^2=3.5$
C. $\sigma^2=4.5$
D. $\sigma^2=5.0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider some function $f(x): \mathbb{R}^d \rightarrow \mathbb{R}$, and assume that we want to run an iterative algorithm to find the maximizer of f. Which update rule should we use to do this (for some $\eta>0$)?

A. $x_{t+1} \leftarrow -x_t+\eta \cdot \nabla_x f\left(x_t\right)$.
B. $x_{t+1} \leftarrow x_t-\eta \cdot \nabla_x f\left(x_t\right)$.
C. $x_{t+1} \leftarrow -x_t-\eta \cdot \nabla_x f\left(x_t\right)$.
D. $x_{t+1} \leftarrow x_t+\eta \cdot \nabla_x f\left(x_t\right)$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Which of the following is/are true about a convex function $f(x) : \mathbb{R}^d \to \mathbb{R}$?

A. $f(x)$ must be differentiable across its entire domain.
B. $f(x)$ has a unique global minimum.
C. $g(x) = -f(x)$ is also convex.
D. If $f(x)$ is twice differentiable, then $z^\top \nabla^2 f(x) z \geq 0$ for all $z \in \mathbb{R}^d$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose the random variable U has uniform distribution on $[0, 1]$ and $X = -2 \log U$. The density of X is

A. $f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
B. $f(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
C. $f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
D. $f(x) = \begin{cases} \frac{x}{2} & \text{if } x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let X and Y be jointly distributed random variables with the joint probability density function (PDF):

$$f(x, y) = \begin{cases} c & \text{if } x^2 + y^2 \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

where $c > 0$.
What is the value of $P(Y > |X|)$?
Hint: Use a geometrical approach instead of solving integrals.

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{1}{6}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Others

A random variable X has probability density function $f(x)$ as given below:

$$f(x) = \begin{cases} ax + b, & \text{for } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If the expected value $E[X] = 2/3$, then $P[X < 0.5]$ is ___

Your Answer: | Correct Answer: 0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Given Set $A = 2, 3, 4, 5$ and Set $B = 11, 12, 13, 14, 15$, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16 ?

A. 0.20

B. 0.25

C. 0.30

D. 0.33

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$.The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$

B. $\frac{9}{16}$

C. $\frac{1}{4}$

D. $\frac{2}{3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Imagine a world where each person has b friends. Alice and Bob are d "friendship links" away, meaning that if $d = 1$, Alice and Bob are directly friends; if $d = 2$, there exists a third person X such that Alice and X are friends, and X and Bob are friends; and so on. An iterative deepening algorithm starts at Alice and aims to find Bob while accessing these friendship links.
In the worst case, how many people does the algorithm need to visit (including Alice and Bob)?

A. b^d

B. $(b-1)d$

C. $\frac{b^{d+1}-1}{b-1}$

D. b^{d+1}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider four sets of vectors A, B, C and D in 3 . Out of these sets of vectors, one set is linearly dependent in 3.
Identity this linearly dependent set.

A. $A = \{[1,1,1],[1,1,0],[1,0,0]\}$
B. $B = \{[1,0,0],[0,1,1],[1,0,1]\}$
C. $C = \{\{1,1,3],[0,2,1],[0,0,0]\}$
D. $D = \{[1,1,1],[0,1,1],[2,0,1]\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

If x is a real number, what are the possible values for the rank of the matrix

$$\begin{bmatrix} 1 & 1 & x \\ 1 & x & 0 \\ x & 0 & 0 \end{bmatrix}?$$

A. 1,2
B. 2,3
C. $0, 1, 2, 3$
D. $0, 1, 2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & -2 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ then b_{32} is:

A. -2
B. -1
C. 0
D. 1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Two matrices A and B are called similar if there exists another matrix S such that $S^{-1}AS = B$.
Consider the statements:

I. If A and B are similar then they have identical rank.
II. If A and B are similar then they have identical trace.
III. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ are similar.

Which of the following is TRUE.

A. Only I
B. Only II
C. Only III
D. All of I, II and III.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which among the following subsets of $\mathbb{R}^3$ is a subspace?

1. $\left\{ \begin{bmatrix} t \\ s \\ \sin t \end{bmatrix} \middle| \, t, s \in \mathbb{R} \right\}$

2. $\left\{ \begin{bmatrix} t \\ 2t \\ 1 \end{bmatrix} \middle| \, t \in \mathbb{R} \right\}$

3. $\left\{ \begin{bmatrix} t \\ s \\ t+s \end{bmatrix} \middle| \, t \in \mathbb{R}, s \geq 0 \right\}$

4. $\left\{ \begin{bmatrix} t \\ t+s \\ s \end{bmatrix} \middle| \, t, s \in \mathbb{R} \right\}$

A. 3 and 4 only
B. 4 only
C. 1, 3 and 4 only
D. All of them

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following describes the least-squares solutions of the equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ -1 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

A. $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ only

B. $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$ only

C. No solutions

D. Infinitely many solutions

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a graph G. Let T be a BFS tree with root r.

Let $d(r, x)$ denote the distance from root node r to node x in T.

If v is visited before u in the breadth first search traversal, which of the following statements is TRUE?

A. $d(r, v) > d(r, u)$
B. $d(r, v) = d(r, u)$
C. $d(r, v) < d(r, u)$
D. $d(r, v) \leq d(r, u)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following option is correct regarding the given statement?
On a connected, undirected graph, Breadth-First Search (BFS) and Depth-First Search (DFS) will produce the same tree if and only if the graph is a tree.

A. True, because BFS and DFS always produce identical trees on any connected undirected graph.
B. True, because BFS and DFS produce the same tree only when the graph is a tree.
C. False, because BFS and DFS can produce the same tree even when the graph contains cycles.
D. False, because BFS and DFS never produce the same tree, even if the graph is a tree.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

In the quicksort algorithm, you want to ensure stability while pivoting around the rightmost item p. Which of the following partitioning strategies should be used?

A. Partition such that the left side of the pivot contains elements $\leq p$, and the right side contains elements > p.
B. Partition such that the left side of the pivot contains elements $<$p, and the right sde contains elements $\geq$p.
C. Partition such that the left side of the pivot contains elements $<$ p, and the right side contains elements $>$ p.
D. It is not possible to make Quicksort a stable sorting algorithm.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $f(n) = \Omega(n), g(n) = O(n)$ and $h(n) = \Theta(n)$. Then $[f(n).\, g(n)] + h(n)$ is:

A. $\Omega(\mathrm{n})$
B. $O(n)$
C. $\theta(n)$
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a binary search tree whose labels are each one capital letter, and assume that a level-order traversal yields the sequence "GENIUS". What is the preorder traversal of this tree?

A. GENIUS
B. GNEUIS
C. GUINES
D. GIUSEN

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose a directed graph has a path from vertex X to vertex Y , but no path from vertex Y to vertex X The relationship between the finish times for depth-first search is:

A. finish $(\mathrm{X}) <$ finish (Y)
B. finish $(\mathrm{X}) >$ finish (Y)
C. $\mathrm{finish}(X) = \mathrm{finish}(Y)$
D. could be either A. or B.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Which of the following is/are TRUE related to schemas used in warehousing databases?

A. In a star schema, the primary keys are stored in the fact table and the foreign keys are stored in the dimension table.
B. Star schemas can help eliminate update errors by reducing duplication of data.
C. There is only one fact table being used in a star schema.
D. There is only one fact table being used in a snowflake schema.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Assume we have access to data $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. Assume that we also have access to weights $\{w_i\}_{i=1}^n, w_i \in \mathbb{R}$ and $w_i > 0$, which gives the "importance" of each data point. Our goal is to solve the weighted least squares regression problem:

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^d} \sum_{i=1}^n w_i \left(x_i^\top \theta - y_i\right)^2.$$

$$\text{Denote } X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \text{ and } W = \begin{bmatrix} w_1 & 0 & \dots & 0 \\ 0 & w_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & w_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

What is $\hat{\theta}$ in terms of X, Y, and W ? One Answer

A. $W\left(X^\top X\right)^{-1} X^\top Y$
B. $\left(X^\top X\right)^{-1} X^\top W Y$
C. $\left(X^\top W X\right)^{-1} X^\top Y$
D. $\left(X^\top W X\right)^{-1} X^\top W Y$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

In the context of logistic regression with the model $P(Y = 1 \mid X) = \sigma\left(X^T \beta\right)$, where Y is a binary variable and X is a vector of features, the following statements are given.
Which of the following statements is/are SOMETIMES true but not ALWAYS?

A. $P(Y = 1 \mid X) > X^T \beta$
B. $P(Y = 1 \mid X) = P(Y = 0 \mid -X)$
C. $P(Y = 1 \mid X) < 1$
D. $\sigma\left(X^T \beta\right) \leq \sigma\left(X^T (2 \cdot \beta)\right)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

In a classification task, the output labels are divided into the following categories:

- 5% of instances belong to Class 1
- 25% belong to Class 2
- 60% belong to Class 3
- 10% belong to Class 4

The accuracy score is calculated as:

$$\text{Average accuracy } = \frac{\sum_{i=1}^N \text{ equals } \left(y_i, \hat{y}_i\right)}{N}$$

where equals $\left(y_i, \hat{y}_i\right) := 1$ when y_i (actual tag) equals $\hat{y}_i$ (predicted tag), and 0 otherwise.
If the model assigns random labels so that each label is equally probable, what would the average accuracy score be?

A. 5%
B. 25%
C. 33.3%
D. 50%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider a centered design matrix $X \in \mathbb{R}^{n \times d}$, where n is the number of examples and d is the dimension of the examples. The singular value decomposition (SVD) of X is given by $X = UDV^\top$. Which of the following matrices provide the coordinates of the data after projecting onto the principal components? Select all that apply.

A. XV
B. UD
C. DV^T
D. XU

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider 30 evenly spaced points (1 to 30) in 1D and 10 clusters. The current cluster assignments are: Points $1, 2, 3$ belong to cluster 1; Points 4, 5,6 belong to cluster 2; and so on, until points $28, 29, 30$ belong to cluster 10. The current cluster centers are $c_1 = 1, c_2 = 4, c_3 = 7, \ldots, c_{10} = 28$. After running k-means, what is the reduction in distortion (K-means loss)?

A. 10
B. 20
C. 30
D. 40

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Suppose we train a hard-margin linear SVM on $n > 100$ data points in R^2, yielding a hyperplane with exactly 2 support vectors. If we add one more data point and retrain the classifier, what is the maximum possible number of support vectors for the new hyperplane (assuming the $n + 1$ points are linearly separable)? Choose the correct answer:-

A. 2
B. n
C. 3
D. $n + 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

You train an SVM on a dataset with data of x_i and labels $y_i = \pm 1$ This produces a vector of weights w and a bias b. For the first 6 training examples, the values of $w^T x_i + b$ are listed below. How many support vectors are there?

i	y_i	$w^T x_i + b$
1	-1	-1.5
2	-1	-1
3	-1	0.5
4	+1	1
5	+1	3
6	-1	4

A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

The probability density function of the random variable X is

$$f(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

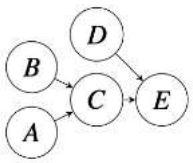
where $\lambda > 0$. For testing the hypothesis $H_0 : \lambda = 3$ against $H_A : \lambda = 5$, a test is given as "Reject H_0 if $X \geq 4.5$ ". The probability of type I error and power of this test are, respectively,

A. 0.1353 and 0.4966
B. 0.1827 and 0.379
C. 0.2021 and 0.4493
D. 0.2231 and 0.4066

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the graphical model displayed below



Which of the following independence statements does not always hold true?

A. A and B are marginally independent.
B. B and E are conditionally independent given C .
C. A and D are marginally independent.
D. A and B are conditionally independent given C .

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider two independent random variables X and Y with identical distributions.The variables X and Y take value 0,1 and 2 with probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$ respectively. What is conditional probability $P(X+Y=2 \mid X-Y=0)$?

A. 0
B. $\frac{1}{16}$
C. $\frac{1}{6}$
D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x:

A. $F(x) - G(x) \leq 0$
B. $F(x) - G(x) \geq 0$
C. $(F(x) - G(x)) \cdot x \leq 0$
D. $(F(x) - G(x)) \cdot x \geq 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

You are taking a multiple-choice exam that has 4 answers for each question. You are a smart student, so in answering a question on this exam, the probability that you know the correct answer is p, and you always choose the correct answer when you know it. If you don't know the answer, you choose one (uniformly) at random. What is the probability that you knew the correct answer to a question, given that you answered it correctly?

A. $\frac{p + \frac{1-p}{4}}{p}$
B. $\frac{p}{1+p}$
C. $\frac{p}{p + \frac{1-p}{4}}$
D. $\frac{p}{\frac{p}{4} + 1}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Suppose that X is a random variable for which $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$, and let c be an arbitrary constant. Which one of these statements is true:

A. $E\left[(X-c)^2\right] = (\mu - c)^2 + \sigma^2$
B. $E\left[(X-c)^2\right] = (\mu - c)^2$
C. $E\left[(X-c)^2\right] = (\mu - c)^2 - \sigma^2$
D. $E\left[(X-c)^2\right] = (\mu - c)^2 + 2\sigma^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Which of the following statements are TRUE regarding positive semi-definite and positive-definite matrices?

A. The singular values of a positive semi-definite matrix are the same as its eigenvalues.
B. If a matrix A is positive semi-definite, then there exists a matrix B such that $B^T B = A$.
C. The covariance matrix of any distribution is positive semi-definite and invertible.
D. If the Hessian of a function is positive semi-definite, then the function is convex.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider the matrix $A \in \mathbb{R}^{3\times 4}$ given by:

$$A = \begin{pmatrix} 1 & 2 & 3 & a \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & b \end{pmatrix}$$

Which of the following statements is/are true regarding the dimension of the leftnull space of A ? A. If $a = 2b$, then $\dim(\text{left null}(A)) = 1$ B. If $a \neq 2b$, then $\dim(\text{left null}(A)) = 0$ C. If $a = 2b$, then $\dim(\text{left null}(A)) = 0$ D If $a \neq 2b$, then $\dim(\text{left null}(A)) = 1$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

You are given the following matrix:

$$A = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 2 & 1 & -3 & 4 \\ 1 & -2 & 1 & -4 \end{pmatrix}$$

For the equation $A^T y = d$ to be solvable, it must hold that $d^T z = 0$ for some vector z. Which of the following is a valid vector z ?

A. $z = (1, 1, 1, 0)$
B. $z = (0, 0, 0, 1)$
C. $z = (2, 2, 2, 1)$
D. $z = (0, 1, 1, 1)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let

$$V = \text{span}\left\{\begin{pmatrix}1\\2\\3\\4\end{pmatrix}, \begin{pmatrix}0\\4\\1\\-1\end{pmatrix}\right\}, \quad W = \text{span}\left\{\begin{pmatrix}4\\3\\1\\-6\end{pmatrix}, \begin{pmatrix}2\\1\\2\\1\end{pmatrix}\right\}$$

be subspaces of $\mathbb{R}^4$. Which of the following is a nonzero vector in $V \cap W$, i.e., a vector that lies in both subspaces?

A. $\begin{pmatrix}2\\0\\5\\9\end{pmatrix}$

B. $\begin{pmatrix}1\\2\\3\\4\end{pmatrix}$

C. $\begin{pmatrix}0\\1\\1\\0\end{pmatrix}$

D. $\begin{pmatrix}4\\3\\1\\-6\end{pmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Let flip(p) be a procedure which returns 1 with probability p and 0 with probability $1 - p$.
Each call to flip is independent.
Consider the two following functions:

```
iterativeF(n,p):
    tot = 0
    for i = i to n:
        tot = tot + flip(p)
    return tot
```

```
recursiveF(n,p):
    if n <= 1:
        return flip(p)
    return 2*recursiveF(n/2,p)
```

Suppose we're given some power of two $n = 2^k$ for $k \in \mathbb{N}$ and a legal probability $p \in [0, 1]$.

A. Expected value of iterative$\mathrm{F}(n, p)$ is np
B. Expected value of iterative$\mathrm{F}(n, p)$ is n^2p
C. Expected value of recursive$\mathrm{F}(n, p)$ is np
D. Expected value of recursive$\mathrm{F}(n, p)$ is 2^{np}

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the scenario where we aim to design a stack with the following operations:

- push(x): Add element x to the stack.
- pop(): Remove the top element from the stack.
- getMinimum(): Retrieve the minimum value from the stack without removing it.

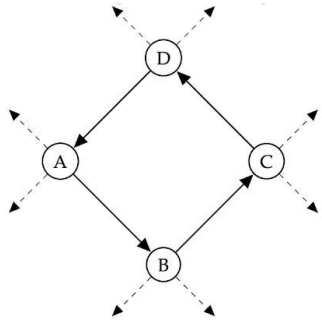
We have the option to utilize additional stack(s) to achieve the desired functionalities. Which of the following statements is true?

A. A minimum of 2 extra stacks is required to implement all three functions (push, pop, and getMinimum) in $O(1)$ time complexity.
B. A minimum of 1 extra stack is required to implement all three functions (push, pop, and getMinimum) in $O(1)$ time complexity.
C. To implement all three functions (push, pop, and getMinimum) in $O(1)$ time complexity, at least one extra stack and one extra queue are needed.
D. Regardless of the number of extra stacks used, it is not possible to implement the getMinimum() function in $\mathrm{O}(1)$ time complexity.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A part of a directed graph is shown below. There are other vertices and edges in the graph that are not shown in the picture (but all edges related to vertices A, B, C, D are shown).
There is no difference between dashed and solid edges; the design is just there to make it look nice.



Consider a DFS run on the entire graph. Indicate whether each of the following subsets of post-values (finish times) is possible or impossible.

Table 1

Vertex v	post $[v]$
A	22
B	100
C	64
D	36

Table 2

Vertex v	post $[v]$
A	22
B	36
C	64
D	100

A. Table 1 is possible but Table 2 is impossible
B. Table 2 is possible but Table 1 is impossible
C. Both Tables are possible
D. Both Tables are impossible

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Numerical Type | Award: 2 | Penalty: 0 | Others

Consider the following hash table, where keys are stored using double hashing. Double hashing is defined as $h = h_1(\text{ key }) + i \cdot h_2(\text{ key })$, where $h_1(\text{ key }) = \text{key} \,\%11$ and $h_2(\text{ key }) = 1 + (\text{ key } \%10)$

0	220
1	660
2	442
3	333
4	
5	555
6	
7	777
8	882
9	999
10	

Now, suppose 222 is to be inserted using double hashing. Which slot will be used?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Which of the following statements is always true?

I. If two matrices are similar, then they have the same determinant.
II. If two matrices have the same characteristic polynomial, then they are similar.
III. If a matrix is diagonalizable, then it is invertible.
IV. If a matrix A is invertible, then zero is not an eigenvalue of A.

A. III and IV only
B. I only
C. I and IV only
D. all of them

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

A relational algebra is defined over sets of tuples.
The relational union of two sets (relations) R and S requires the schemes of R and S to be union-compatible i.e. to share attribute names $\&$ their respective domains, and the order of attributes to be the same for both relations.
In the relational algebra, two union-compatible sets with P and Q records (table lengths) are joined by a binary operator. What is the minimum and maximum number of records in the answer if the operation is union? What if it is a natural join? (Let $P <= Q$)

A. For union operation, minimum is P, maximum is $P + Q$.
B. For union operation, minimum is Q, maximum is $P + Q$.
C. For natural join operation, minimum is 0 , maximum is PQ.
D. For natural join operation, minimum is 0 , maximum is P.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Which of the following is/are NOT a valid rule about functional dependencies (FD's) in the context of RDBMS model?

A. Reversitivity: if "A $\rightarrow$ B " then " $\mathrm{B} \rightarrow \mathrm{A}$ "
B. Kindatransitivity: if "A $\rightarrow$ C " and " $\mathrm{AB} \rightarrow \mathrm{C}$ " then " $\mathrm{B} \rightarrow \mathrm{C}$ "
C. Pseudotransitivity: if "A $\rightarrow$ B " and " $\mathrm{BC} \rightarrow \mathrm{D}$ " then " $\mathrm{AC} \rightarrow \mathrm{D}$ "
D. Union on LHS: If " $A \rightarrow B$ " and and " $C \rightarrow B$ " then " $AC \rightarrow B$ "

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A binary operator is defined as follows

$$P \Updownarrow Q = \sim P \wedge Q$$

Which of the following statement is equivalent to $P \rightarrow Q$

A. $\sim P \Updownarrow Q$
B. $\sim (P \Updownarrow Q)$
C. $\sim (\sim P \Updownarrow Q)$
D. $\sim (\sim P \Updownarrow \sim Q)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider the following predicates:

- $\mathrm{bird}(x)$: x is a bird.
- $\mathrm{penguin}(x)$: x is a penguin.
- $\mathrm{fly}(x)$: x can fly.

Consider the following sentence:

All birds except penguins' fly. (Note that Penguins may or may not fly)

Which of the following is/are a formula in first-order logic expressing the given fact?

A. $\forall x.\left[(\mathrm{bird}(x) \wedge \neg \text{ penguin } (x)) \rightarrow \mathrm{fly}(x)\right]$
B. $\forall x.\left[\neg \mathrm{bird}(x) \vee \neg \text{ penguin } (x) \vee \text{ fly } (x)\right]$
C. $\forall x.\left[\neg \text{ bird } (x) \vee \text{ penguin } (x) \vee \text{ fly } (x)\right]$
D. $\forall x.\left[\mathrm{bird}(x) \vee \text{ penguin } (x) \vee \text{ fly } (x)\right]$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Others

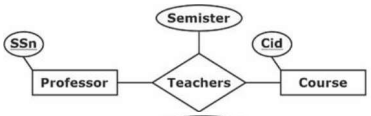
The order of a leaf node in a B tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is $1\ \mathrm{K}$ bytes(i.e. 1024 bytes), data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

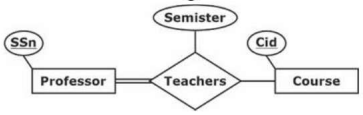
Your Answer: | Correct Answer: 46 | Not Attempted | Time taken: 00min 00sec | Discuss

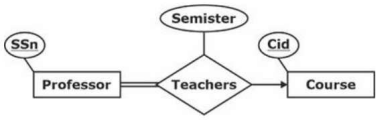
Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

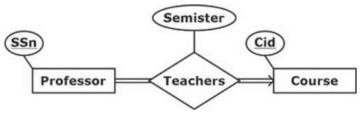
Which of the following ER diagram represents the given requirements

"Every professor teaches exactly one course"

A.


B.


C.


D.


Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

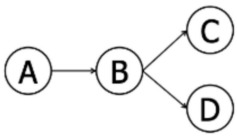
Let $\mathrm{nbs}(m) =$ the number of possible Bayes Network graph structures using m attributes. (Note that two networks with the same structure but different probabilities in their tables do not count as different structures). Which of the following statements is true?

A. $m \leq \mathrm{nbs}(m) < \frac{m(m-1)}{2}$
B. $\frac{m(m-1)}{2} \leq \mathrm{nbs}(m) < 2^m$
C. $2^m \leq \mathrm{nbs}(m) < 2^{\frac{m(m-1)}{2}}$
D. $2^{\frac{m(m-1)}{2}} \leq \mathrm{nbs}(m)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Others

Assume the following Bayes net, and the corresponding distributions over the variables in the Bayes net:



A	$P(A)$
$+a$	$1/5$
$-a$	$4/5$

B	C	$P(C \mid B)$
$+b$	$+c$	$1/4$
$+b$	$-c$	$3/4$
$-b$	$+c$	$2/5$
$-b$	$-c$	$3/5$

A	B	$P(B \mid A)$
$+a$	$+b$	$1/5$
$+a$	$-b$	$4/5$
$-a$	$+b$	$1/2$
$-a$	$-b$	$1/2$

B	D	$P(D \mid B)$
$+b$	$+d$	$1/2$
$+b$	$-d$	$1/2$
$-b$	$+d$	$4/5$
$-b$	$-d$	$1/5$

Your task is now to estimate $P(+b \mid -a, -c, -d)$ using rejection sampling. Below are some samples that have been produced by prior sampling (that is, the rejection stage in rejection sampling hasn't happened yet).

$$\begin{array}{cccc} -a & -b & +c & +d \\ +a & -b & -c & +d \\ -a & -b & +c & -d \end{array} \qquad \begin{array}{cccc} -a & -b & -c & -d \\ -a & +b & +c & +d \\ +a & -b & -c & -d \end{array}$$

Using those samples, what value would you estimate for $P(+b \mid -a, -c, -d)$ using rejection sampling?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

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